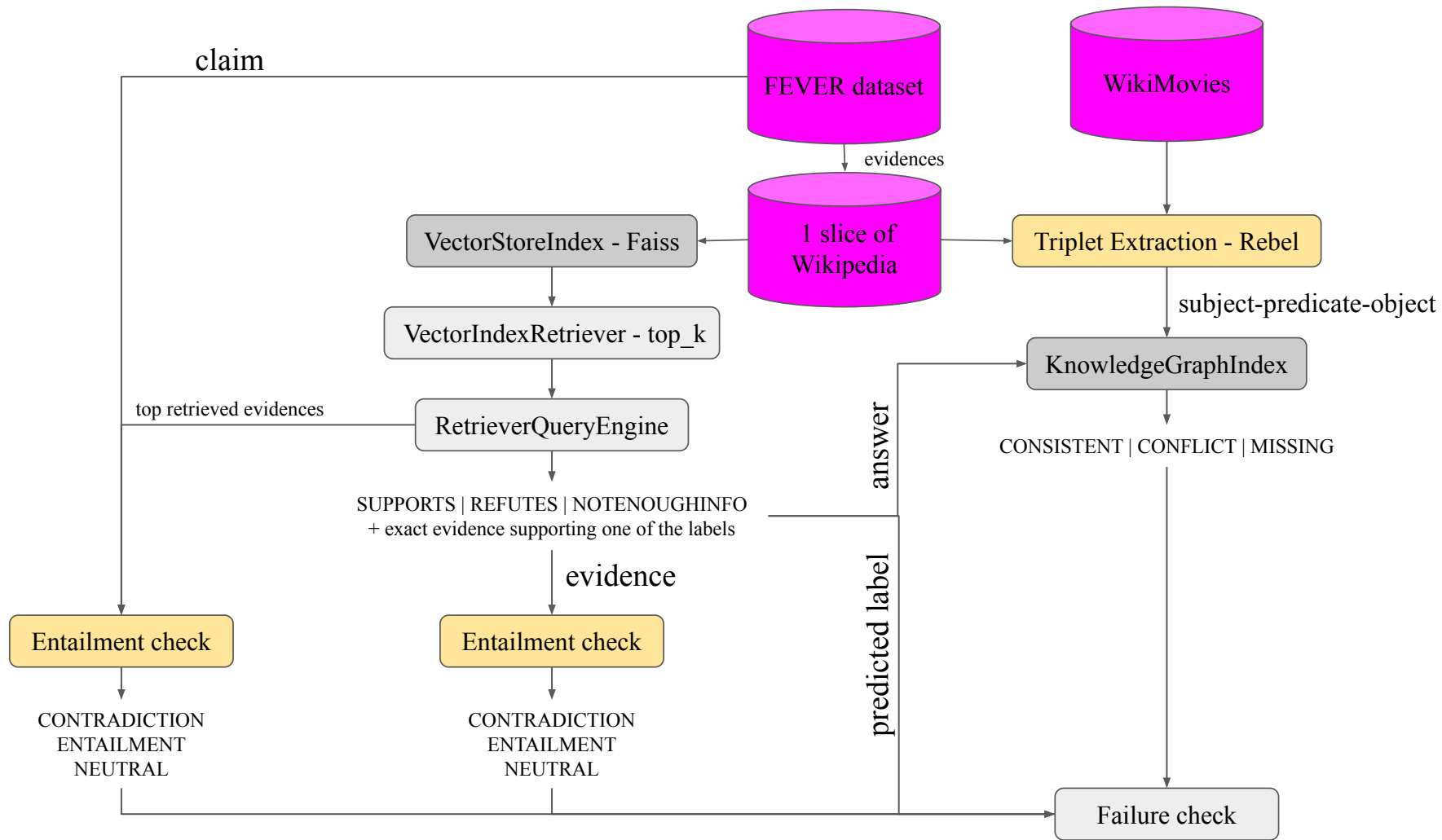
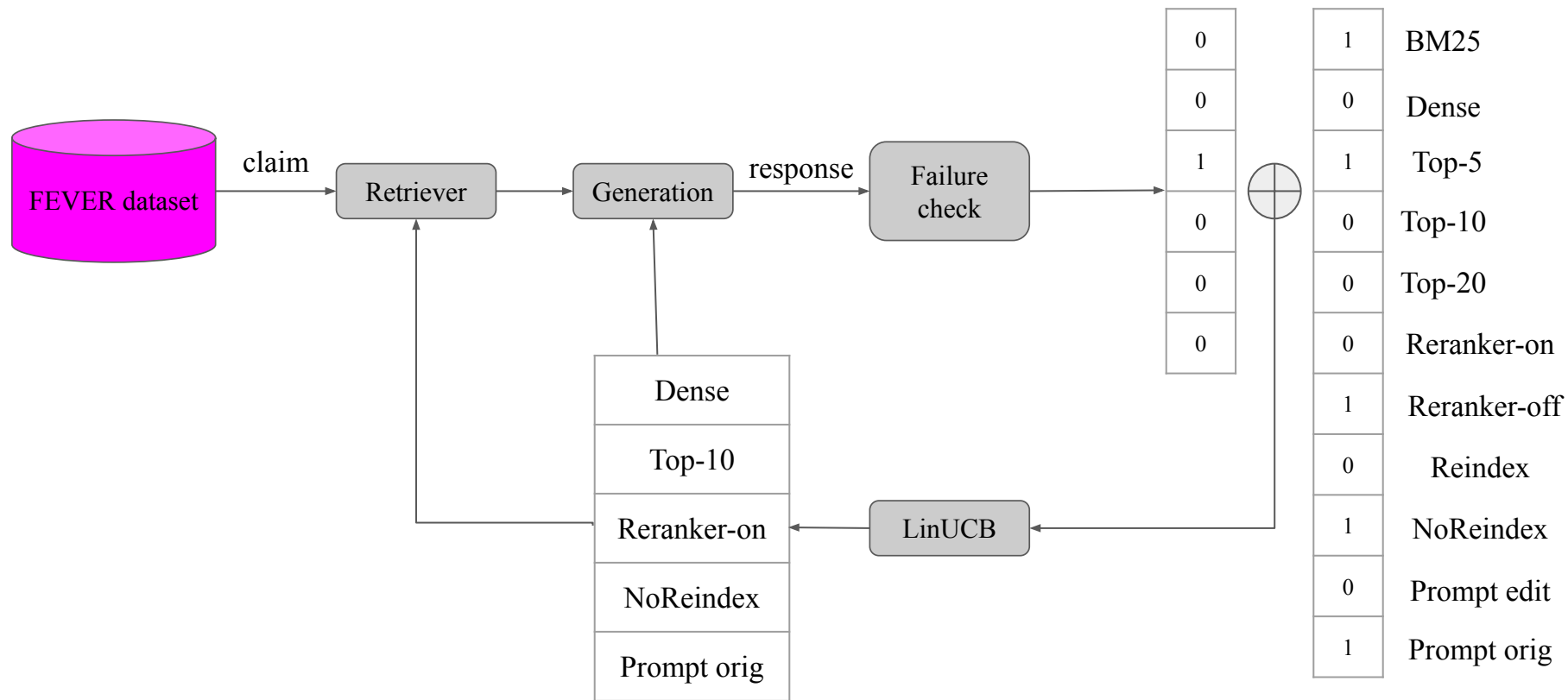




Failure check:

- KG check is missing:
 - flag = *MISSING_ENTITY*
- KG check is conflict:
 - flag = *MISSING_PREDICATE*
- RAG label supports + claim entails:
 - flag = *SUCCESSFUL*
- RAG label refutes + claim contradicts:
 - flag = *SUCCESSFUL*
- RAG label is NEI + claim is neutral:
 - flag = *SUCCESSFUL*
- Otherwise:
 - claim entails: flag = *WRONG_LABEL_SUPPORTS*
 - claim contradicts: flag = *WRONG_LABEL_REFUTES*
 - claim is neutral: flag = *WRONG_LABEL_NOTENOUGHINFO*



LinUCB training process:

- Context

- Failure check, one-hot vector
- Default hyperparameters, one-hot vector

- Action

- Action = a combination of hyperparameters
- Each combination associated with a unique action_id
- Transformed to one-hot vector

- Reward

- $10 * \text{failure check successful} +$
- $5 * (\text{WRONG_LABEL_SUPPORTS} + \text{WRONG_LABEL_REFUTES} + \text{WRONG_LABEL_NOTENOUGHINFO}) +$
- $0 * (\text{MISSING_ENTITY} + \text{MISSING_PREDICATE}) +$
- $10 * [\text{RAG predicted label} == \text{FEVER label}] +$
- - latency / budget
- - vram / budget

